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Predicting drug responses of unseen cell types through transfer learning with foundation models

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Abstract

Drug repurposing through single-cell perturbation response prediction provides a cost-effective approach for drug development, but accurately predicting responses in unseen cell types that emerge during disease progression remains challenging. Existing methods struggle to achieve generalizable cell-type-specific predictions. To address these limitations, we introduce the cell-type-specific drug perturbatIon responses predictor (CRISP), a framework for predicting perturbation responses in previously unseen cell types at single-cell resolution. CRISP leverages foundation models and cell-type-specific learning strategies to enable effective transfer of information from control to perturbed states even with limited empirical data. Through systematic evaluation across increasingly challenging scenarios, from unseen cell types to cross-platform predictions, CRISP shows generalizability and performance improvements. We demonstrate CRISP's drug repurposing potential through zero-shot prediction from solid tumor data to sorafenib's therapeutic effects in chronic myeloid leukemia. The predicted anti-tumor mechanisms, including CXCR4 pathway inhibition, are supported by independent studies as an effective therapeutic strategy in chronic myeloid leukemia, aligning with past studies and clinical trials.

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Data availability

All of the datasets used in this research can be accessed publicly. The single-cell drug perturbation dataset SciPlex3[2] and the GBM dataset[47] can be obtained via GEO using access codes GSM4150378 and GSE148842. The NeurIPS dataset[27] was obtained from NeurIPS competition; the PC9 dataset[48] can be obtained via GEO with access code GSE149383. The unperturbed PBMC-Bench data[49] can be found here. Detailed description of these datasets can be found in the 'Datasets' section of the Methods. Source data are provided with this paper.

Code availability

The open-source codes of CRISP can be found via https://github.com/ml4bio/CRISP and on Zenodo via https://doi.org/10.5281/zenodo.16713259 (ref. [56]).

References

1. Van de Sande, B. et al. Applications of single-cell RNA sequencing in drug discovery and development. *Nat. Rev. Drug Discov.* **22**, 496–520 (2023).

2. Srivatsan, S. R. et al. Massively multiplex chemical transcriptomics at single-cell resolution. *Science* **367**, 45–51 (2020).

3. Zang, C. et al. High-throughput target trial emulation for Alzheimer's disease drug repurposing with real-world data. *Nat. Commun.* **14**, 8180 (2023).

4. Rodriguez, S. et al. Machine learning identifies candidates for drug repurposing in Alzheimer's disease. *Nat. Commun.* **12**, 1033 (2021).

5. Charpignon, Marie-Laure et al. Causal inference in medical records and complementary systems pharmacology for metformin drug repurposing towards dementia. *Nat. Commun.* **13**, 7652 (2022).

6. Elhasasna, H. et al. A drug repurposing screen identifies fludarabine phosphate as a potential therapeutic agent for N-MYC overexpressing neuroendocrine prostate cancers. *Cells* **11**, 2246 (2022).

7. Yu, M. et al. Deep learning large-scale drug discovery and repurposing. *Nat. Comput. Sci.* **4**, 600–614 (2024).

8. Pushpakom, S. et al. Drug repurposing: progress, challenges and recommendations. *Nat. Rev. Drug Discov.* **18**, 41–58 (2019).

9. Stubbington, MichaelJ. T., Rozenblatt-Rosen, O., Regev, A. & Teichmann, S. A. Single-cell transcriptomics to explore the immune system in health and disease. *Science* **358**, 58–63 (2017).

10. Papalexi, E. & Satija, R. Single-cell rna sequencing to explore immune cell heterogeneity. *Nat. Rev. Immunol.* **18**, 35–45 (2018).

11. Norman, T. M. et al. Exploring genetic interaction manifolds constructed from rich single-cell phenotypes. *Science* **365**, 786–793 (2019).

12. Replogle, J. M. et al. Mapping information-rich genotype-phenotype landscapes with genome-scale perturb-SEQ. *Cell* **185**, 2559–2575 (2022).

13. Doench, J. G. Am i ready for CRISPR? A user's guide to genetic screens. *Nat. Rev. Genet.* **19**, 67–80 (2018).

14. Nakamura, M., Gao, Y., Dominguez, A. A. & Qi, L. S. CRISPR technologies for precise epigenome editing. *Nat. Cell Biol.* **23**, 11–22 (2021).

15. Hanna, R. E. & Doench, J. G. Design and analysis of crispr–cas experiments. *Nat. Biotechnol.* **38**, 813–823 (2020).

16. He, B. et al. Combination therapeutics in complex diseases. *J. Cell. Mol. Medicine* **20**, 2231–2240 (2016).

17. Jagadeesh, K. A. et al. Identifying disease-critical cell types and cellular processes by integrating single-cell RNA-sequencing and human genetics. *Nat. Genet.* **54**, 1479–1492 (2022).

18. Lotfollahi, M., Wolf, F. A. & Theis, F. J. scGen predicts single-cell perturbation responses. *Nat. Methods* **16**, 715–721 (2019).

19. Bunne, C. et al. Learning single-cell perturbation responses using neural optimal transport. *Nat. Methods* **20**, 1759–1768 (2023).

20. Piran, Z., Cohen, N., Hoshen, Y. & Nitzan, M. Disentanglement of single-cell data with biolord. *Nat. Biotechnol.* **42**, 1678–1683 (2024).

21. Hetzel, L. et al. Predicting cellular responses to novel drug perturbations at a single-cell resolution. In *Proc. 36th International Conference on Neural Information Processing System* Vol. 35, 26711–26722 (2022).

22. Lotfollahi, M. et al. Predicting cellular responses to complex perturbations in high-throughput screens. *Mol. Syst. Biol.* e11517 (2023).

23. Theodoris, C. V. et al. Transfer learning enables predictions in network biology. *Nature* **618**, 616–624 (2023).

24. Cui, H. et al. scGPT: toward building a foundation model for single-cell multi-omics using generative AI. *Nat. Methods* **21**, 1470–1480 (2024).

25. Hao, M. et al. Large-scale foundation model on single-cell transcriptomics. *Nat. Methods* **21**, 1481–1491 (2024).

26. Heimberg, G. et al. A cell atlas foundation model for scalable search of similar human cells. *Nature* **638**, 1085–1094 (2025).

27. *Neurips 2023: Single-cell Perturbation Prediction* (NeurIPS, 2023); https://www.kaggle.com/competitions/open-problems-single-cell-perturbations/data

28. Burger, J. A. & Okkenhaug, K. Idelalisib—targeting PI3Kδ in patients with B-cell malignancies. *Nat. Rev. Clin. Oncol.* **11**, 184–186 (2014).

29. Petroni, G., Formenti, S. C., Chen-Kiang, S. & Galluzzi, L. Immunomodulation by anticancer cell cycle inhibitors. *Nat. Rev. Immunol.* **20**, 669–679 (2020).

30. Wu, X. et al. Recent advances in dual PI3K/mTOR inhibitors for tumour treatment. *Front. Pharmacol.* **13**, 875372 (2022).

31. Shah, N. P. et al. Sequential ABL kinase inhibitor therapy selects for compound drug-resistant BCR-ABL mutations with altered oncogenic potency. *J. Clin. Investigat.* **117**, 2562–2569 (2007).

32. Wilhelm, S. et al. Discovery and development of sorafenib: a multikinase inhibitor for treating cancer. *Nat. Rev. Drug Discov.* **5**, 835–844 (2006).

33. Kanehisa, M. & Goto, S. Kegg: Kyoto encyclopedia of genes and genomes. *Nucleic Acids Res.* **28**, 27–30 (2000).

34. Paech, F. et al. Mechanisms of mitochondrial toxicity of the kinase inhibitors ponatinib, regorafenib and sorafenib in human hepatic HepG2 cells. *Toxicology* **395**, 34–44 (2018).

35. Ek, F. et al. Sorafenib and nitazoxanide disrupt mitochondrial function and inhibit regrowth capacity in three-dimensional models of hepatocellular and colorectal carcinoma. *Sci. Rep.* **12**, 8943 (2022).

36. Weisberg, E. et al. Inhibition of CXCR4 in CML cells disrupts their interaction with the bone marrow microenvironment and sensitizes them to nilotinib. *Leukemia* **26**, 985–990 (2012).

37. *Sorafenib Completed Phase 2 Trials for Chronic Myelogenous Leukemia (CML) Treatment* (DrugBank, accessed 3 August 2025); https://go.drugbank.com/drugs/DB00398/clinical_trials?conditions=DBCOND0040386&phase=2&purpose=treatment&status=completed

38. Lee, E. Q. et al. Phase ii study of panobinostat in combination with bevacizumab for recurrent glioblastoma and anaplastic glioma. *Neuro-oncology* **17**, 862–867 (2015).

39. Xiong, S., Van Pelt, C. S., Elizondo-Fraire, A. C., Liu, G. & Lozano, G. Synergistic roles of Mdm2 and Mdm4 for p53 inhibition in central nervous system development. *Proc. Natl Acad. Sci. USA* **103**, 3226–3231 (2006).

40. Konopleva, M. et al. MDM2 inhibition: an important step forward in cancer therapy. *Leukemia* **34**, 2858–2874 (2020).

41. Scripture, C. D. & Figg, W. D. Drug interactions in cancer therapy. *Nat. Rev. Cancer* **6**, 546–558 (2006).

42. Dong, M. et al. Causal identification of single-cell experimental perturbation effects with CINEMA-OT. *Nat. Methods* **20**, 1769–1779 (2023).

43. Landrum, G. et al. *RDKit: Open-Source Cheminformatics Software* (RDKit, 2016).

44. Knox, C. et al. Drugbank 6.0: the drugbank knowledgebase for 2024. *Nucleic Acids Res.* **52**, D1265–D1275 (2024).

45. Kim, S. et al. Pubchem 2023 update. *Nucleic Acids Res.* **51**, D1373–D1380 (2023).

46. Stathias, V. et al. Lincs data portal 2.0: next generation access point for perturbation-response signatures. *Nucleic Acids Res.* **48**, D431–D439 (2020).

47. Zhao, W. et al. Deconvolution of cell type-specific drug responses in human tumor tissue with single-cell RNA-seq. *Genome Med.* **13**, 82 (2021).

48. Aissa, A. F. et al. Single-cell transcriptional changes associated with drug tolerance and response to combination therapies in cancer. *Nat. Commun.* **12**, 1628 (2021).

49. Ding, J. et al. Systematic comparison of single-cell and single-nucleus RNA-sequencing methods. *Nat. Biotechnol.* **38**, 737–746 (2020).

50. Wolf, F. A., Angerer, P. & Theis, F. J. SCANPY: large-scale single-cell gene expression data analysis. *Genome Biol,.***19**, 15 (2018).

51. Virtanen, P. et al. Scipy 1.0: fundamental algorithms for scientific computing in Python. *Nat. Methods* **17**, 261–272 (2020).

52. Detlefsen, NickiSkafte et al. TorchMetrics-measuring reproducibility in PyTorch. *J. Open Source Softw.* **7**, 4101 (2022).

53. Feydy, J. et al. Interpolating between optimal transport and MMD using sinkhorn divergences. In *Proc. 22nd International Conference on Artificial Intelligence and Statistics* 2681–2690 (PMLR, 2019).

54. Cuturi, M. Sinkhorn distances: lightspeed computation of optimal transport. In *Proc. Advances in Neural Information Processing Systems* Vol. 26 (NIPS, 2013).

55. Fang, Z., Liu, X. & Peltz, G. GSEApy: a comprehensive package for performing gene set enrichment analysis in Python. *Bioinformatics* **39**, btac757 (2023).

56. Wang, Y. et al. Predicting drug responses of unseen cell types through transfer learning with foundation models. *Zenodo* https://doi.org/10.5281/zenodo.16713259 (2025).

57. Crino, L. et al. Initial phase II results with crizotinib in advanced ALK-positive non-small cell lung cancer (NSCLC): profile 1005. *J. Clin. Oncol.* **29**, 7514 (2011).

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Contributions

Y.L. conceived the research. Y.W. led the project and designed the experiments. X.L. ran baseline models, prepared detailed tutorials, and maintained the codebase. Both Y.W. and X.L. contributed to model architecture and algorithm implementation, as well as interpretation of the results. Y.W. and X.L. organized and processed the publicly available data. Y.W. and X.L. visualized the experimental results. Y.L. and I.K. supervised the research. Y.W. and X.L. wrote the paper. All authors read and approved the final paper.

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Ethics declarations

Competing interests

The authors declare no competing interests.

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Extended data

Extended Data Fig. 1 Prediction for cancer cells from the PBMC-trained CRISP.

a. Illustration of cross-tissue prediction. We applied the trained CRISP from the NeurIPS dataset (PBMC) to cancer cells directly and predict the post-perturbation responses to both seen drug and unseen drug. **b**. Scatter plot showing the alignment of predicted log(FC) and observed log(FC), for Crizotinib-treated NSCLC. Crizotinib is a tyrosine kinase inhibitor that has approved for NSCLC treatment[57]. **c**. UMAP visualization of predicted Crizotinib-treated samples, colored by tissues and cell types, respectively. It includes PBMC and cancer cell lines from the NeurIPS, PBMC-Bench, SciPlex3 and PC9 dataset. **d**,**e**. Evaluation results for drug response prediction for NSCLC to the seen and unseen drug. We take CRISP without cell type-specific contrastive learning and classifier as baseline. Although it's not robust enough for unseen drug when predict across tissue, the addition of cell type-specific regularization makes progress. **f**. Comparison between predicted and observed log(FC) on top20 DE genes for Erlotinib-treated NSCLC. Each bar shows the averaged log(FC) on the corresponding gene. It reflects the predicted performance on direction and absolute value. Some icons are created with BioRender.com.

Source data.

Extended Data Fig. 2 Drug screening on SciPlex3 dataset.

a. We perform drug screening on all measured drugs in the Sciplex3 dataset, with the held-out K562 cell line (Chronic Myeloid Leukemia, CML) serving as the target for prediction. A set of CML-related pathways, which are significantly down-regulated upon treatment with approved CML drugs, is selected for screening and ground truth identification. Since nearly all significant pathways are down-regulated, we consider only negatively enriched terms. For real experimental data, a drug is classified as an anti-cancer drug (ground truth) if its significantly enriched down-regulated pathways overlap with the CML-related pathways. For CRISP's predictions, we rank all candidate drugs based on the sum of enrichment scores of CML-related pathways and compute Precision@K using the ground truth anti-cancer drug labels. Additionally, we evaluate the

precision of CRISP's predicted enriched terms by comparing them with real data for each drug. **b**. Precision@K for different values of K in the ranked list of candidate drugs. Notably, the top 10 ranked drugs are all labeled as anti-cancer drugs. **c**. The ranking of enrichment scores for the top 30 candidate drugs. Among them, only four are not anti-cancer drugs (labeled in gray). The approved CML drug 'Bosutinib' ranks 4th in our screening results. **d**. The precision of selected anti-cancer drugs, sorted by precision value. Specifically, the three approved CML drugs (highlighted with red frames)—'Dasatinib,' 'Nilotinib,' and 'Bosutinib'—achieve precision values of approximately 0.7 and rank within the top 30.

Source data.

Supplementary information

Supplementary Information

Supplementary Algorithm, Notes 1 and 2, Figs. 1–8 and Table 1.

Reporting Summary

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Source data

Source Data Fig. 2

Statistical source data.

Source Data Fig. 3

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Source Data Extended Data Fig. 2

Statistical source data.

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